

Data Warehousing and Data Mining – CSA1607

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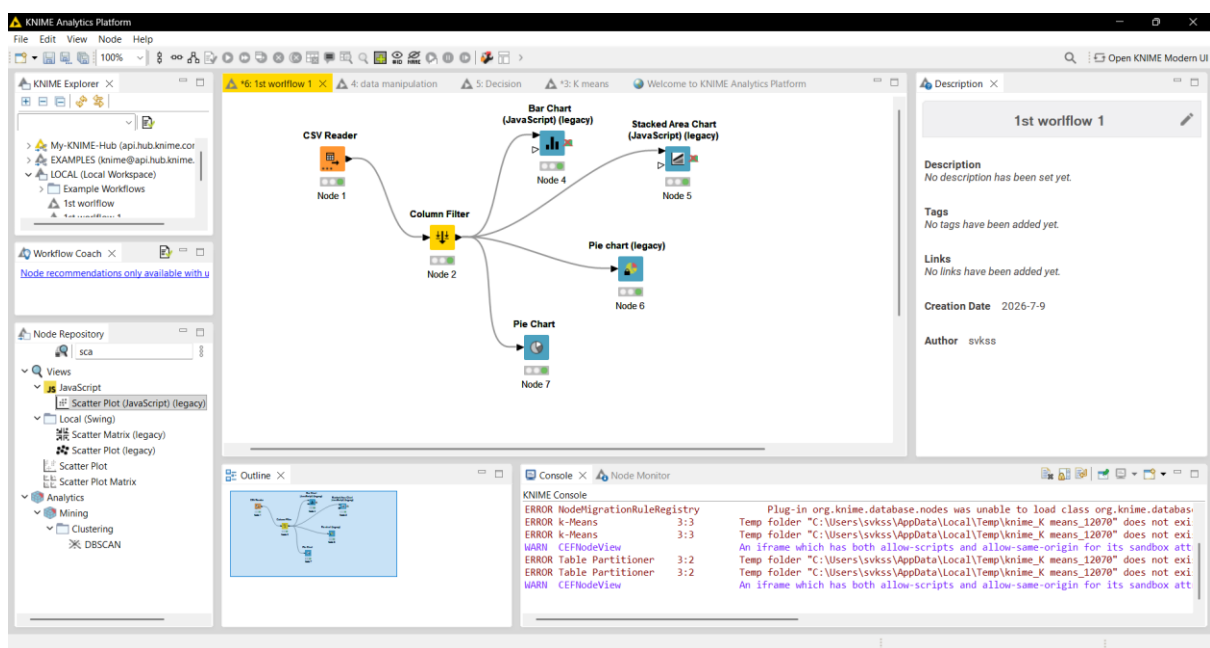
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ACTIVITY 2 – GENERATING REPORT USING KNIME

Objective

To generate a report by documenting different KNIME workflows and their outputs in a structured format for analysis and presentation.

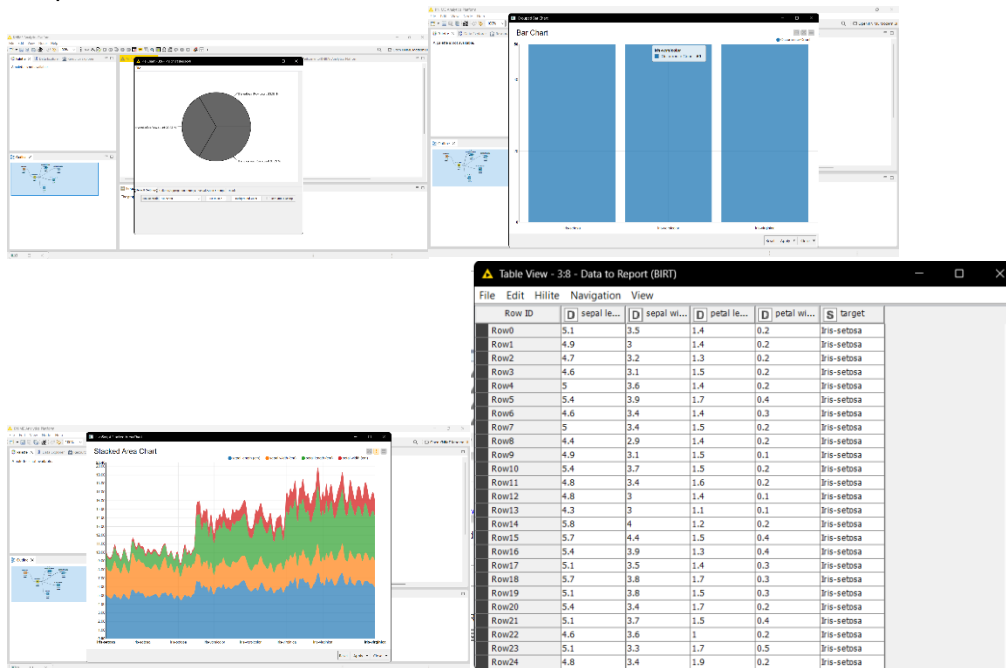
1. Workflow 1



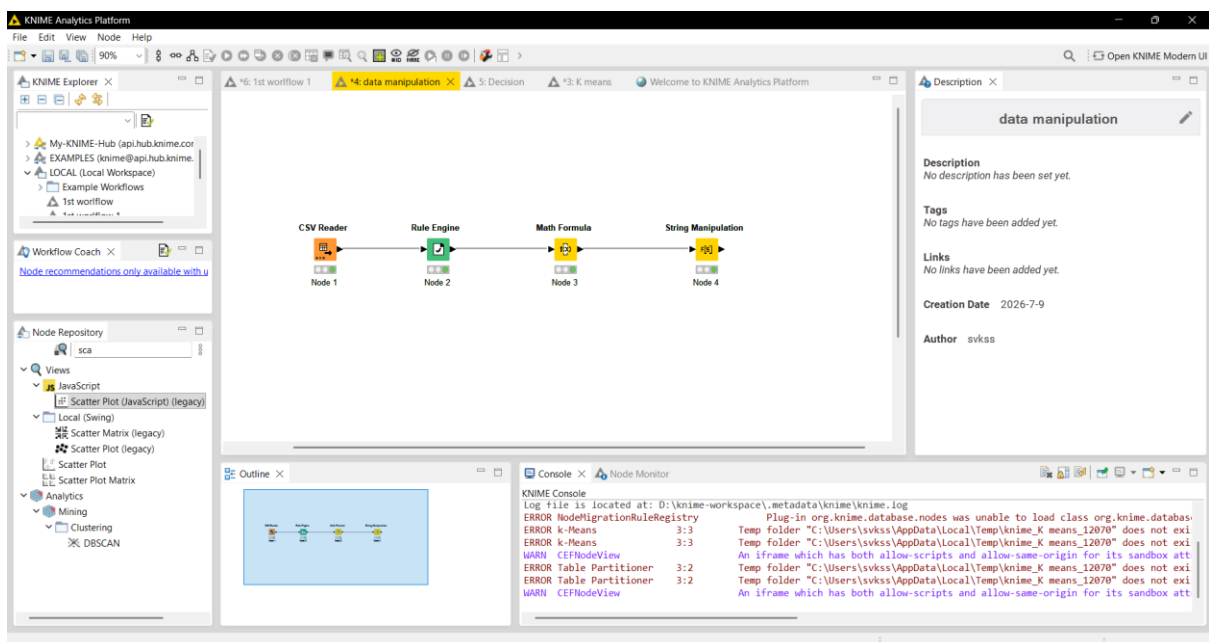
Description

The workflow begins with a CSV Reader node to import the dataset. A Column Filter node is used to select the required columns for analysis. The filtered data is then visualized using Bar Chart, Pie Chart, and Stacked Area Chart nodes to represent the data graphically.

Outputs:



2. Data Manipulation



Description

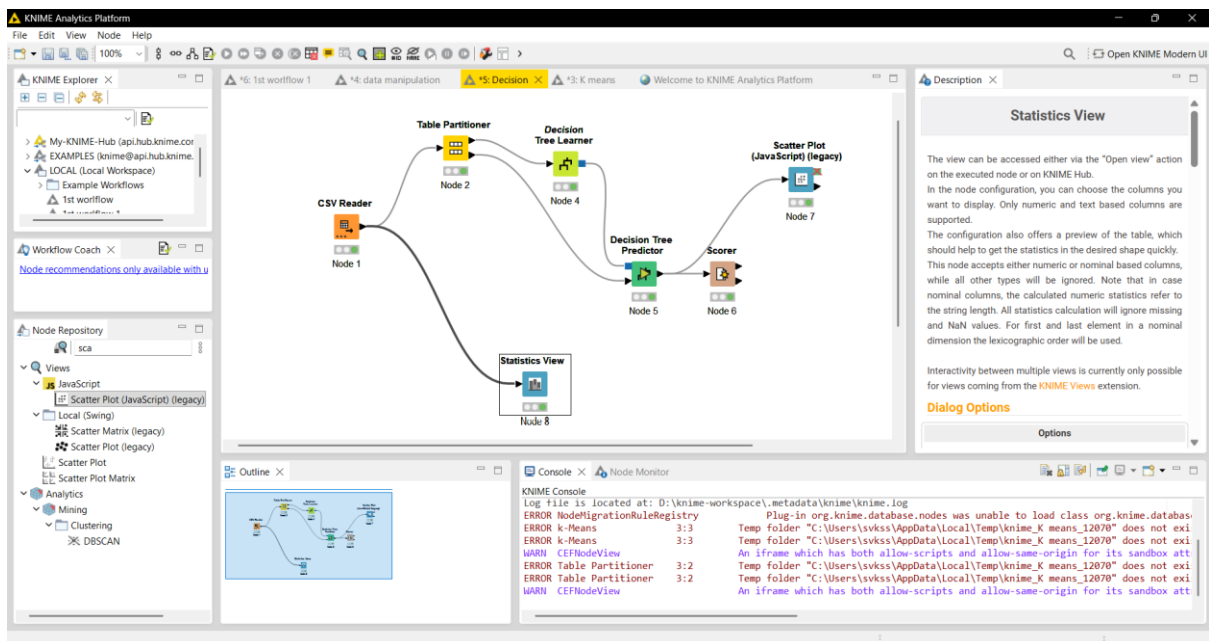
The dataset is imported using the CSV Reader node. The Rule Engine applies conditions to modify or categorize the data. The Math Formula node performs calculations, while the String Manipulation node is used to edit and transform text values. This workflow demonstrates basic data preprocessing techniques.

Outputs:

Table View - 45 - Data to Report (BIRT)

Row ID	sepal le...	sepal ve...	petal le...	petal ve...	target	prediction	new col...	new col...
Row1	5.1	3.5	1.4	0.2	iris-setosa	?	5.2	setosa-est
Row2	4.9	3	1.4	0.2	iris-setosa	?	5.2	setosa-est
Row3	4.7	3.2	1.3	0.2	iris-setosa	?	5.2	setosa-est
Row4	4.6	3.1	1.5	0.2	iris-setosa	?	5.2	setosa-est
Row5	5	3.6	1.4	0.2	iris-setosa	?	5.2	setosa-est
Row6	5.4	3.9	1.7	0.4	iris-setosa	?	5.4	setosa-est
Row7	4.6	3.4	1.4	0.3	iris-setosa	?	5.3	setosa-est
Row8	5	3.4	1.5	0.2	iris-setosa	?	5.2	setosa-est
Row9	4.4	2.9	1.4	0.2	iris-setosa	?	5.2	setosa-est
Row10	4.9	3.1	1.5	0.1	iris-setosa	0.4	5.1	setosa-est
Row11	5.4	3.7	1.5	0.2	iris-setosa	?	5.2	setosa-est
Row12	4.8	3.4	1.6	0.2	iris-setosa	?	5.2	setosa-est
Row13	4.8	3	1.4	0.1	iris-setosa	0.4	5.1	setosa-est
Row14	4.2	3	1.1	0.1	iris-setosa	0.4	5.1	setosa-est
Row15	5.8	4	1.2	0.2	iris-setosa	?	5.2	setosa-est
Row16	5.7	4.4	1.5	0.4	iris-setosa	?	5.4	setosa-est
Row17	5.4	3.9	1.3	0.4	iris-setosa	?	5.4	setosa-est
Row18	5.1	3.5	1.4	0.3	iris-setosa	?	5.3	setosa-est
Row19	5.7	3.8	1.7	0.3	iris-setosa	?	5.3	setosa-est
Row20	5.1	3.8	1.5	0.3	iris-setosa	?	5.3	setosa-est
Row21	5.4	3.4	1.7	0.2	iris-setosa	?	5.2	setosa-est
Row22	5.1	3.7	1.5	0.4	iris-setosa	?	5.4	setosa-est
Row23	4.8	3.8	1	0.2	iris-setosa	?	5.2	setosa-est
Row24	5.1	3.3	1.7	0.5	iris-setosa	?	5.5	setosa-est
Row25	4.8	3.4	1.9	0.2	iris-setosa	?	5.2	setosa-est
Row26	5	3	1.6	0.2	iris-setosa	?	5.2	setosa-est

3. Decision Tree



Description

The dataset is divided into training and testing sets using the Table Partitioner node. A Decision Tree Learner builds the classification model, and the Decision Tree Predictor applies the model to the test data. The Scorer node evaluates the prediction results, while the Statistics View and Scatter Plot help visualize the model's performance.

Outputs:

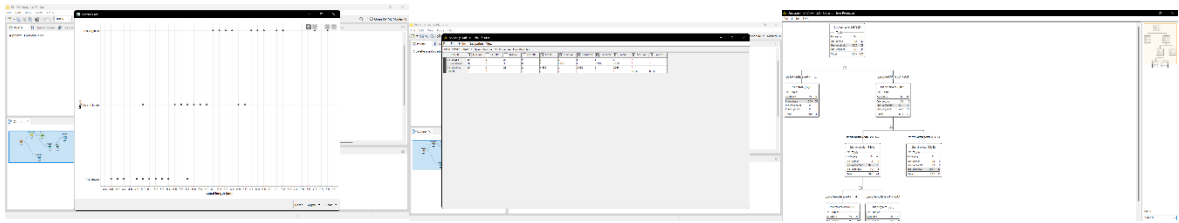
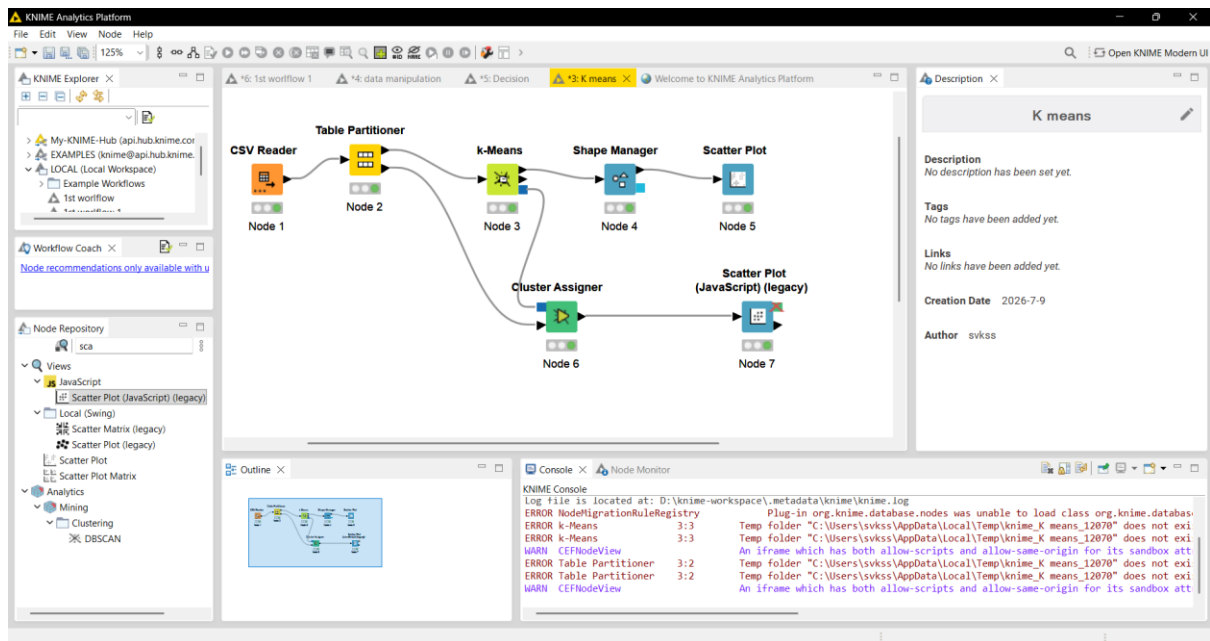


Table View - 5.9 - Data to Report (BIRT)									
File Edit Hilitr Navigation View									
Row ID	D sepal le...	D sepal wi...	D petal le...	D petal wi...	S target	D P (targe...	D P (targe...	D P (	
Row0	5.1	3.5	1.4	0.2	Iris-setosa	1	0	0	
Row1	4.9	3	1.4	0.2	Iris-setosa	1	0	0	
Row4	5	3.6	1.4	0.2	Iris-setosa	1	0	0	
Row6	4.6	3.4	1.4	0.3	Iris-setosa	1	0	0	
Row9	4.9	3.1	1.5	0.1	Iris-setosa	1	0	0	
Row16	5.4	3.9	1.3	0.4	Iris-setosa	1	0	0	
Row18	5.7	3.8	1.7	0.3	Iris-setosa	1	0	0	
Row19	5.1	3.8	1.3	0.3	Iris-setosa	1	0	0	
Row21	5.1	3.7	1.5	0.4	Iris-setosa	1	0	0	
Row22	4.6	3.6	1	0.2	Iris-setosa	1	0	0	
Row23	5.1	3.3	1.7	0.5	Iris-setosa	1	0	0	
Row27	5.2	3.5	1.3	0.2	Iris-setosa	1	0	0	
Row29	4.7	3.2	1.6	0.2	Iris-setosa	1	0	0	
Row31	5.4	3.4	1.5	0.4	Iris-setosa	1	0	0	
Row35	5	3.2	1.2	0.2	Iris-setosa	1	0	0	
Row41	4.5	2.3	1.3	0.3	Iris-setosa	1	0	0	
Row44	5.1	3.8	1.9	0.4	Iris-setosa	1	0	0	
Row48	5.3	3.7	1.5	0.2	Iris-setosa	1	0	0	
Row53	5.5	2.3	4	1.3	Iris-versicolor	0	0.974	0.026	
Row54	6.5	2.8	4.6	1.5	Iris-versicolor	0	0.974	0.026	
Row58	6.6	2.9	4.6	1.3	Iris-versicolor	0	0.974	0.026	
Row60	5	2	3.5	1	Iris-versicolor	0	0.974	0.026	
Row64	5.6	2.9	3.6	1.3	Iris-versicolor	0	0.974	0.026	
Row66	5.6	3	4.5	1.5	Iris-versicolor	0	0.974	0.026	
Row67	5.8	2.7	4.1	1	Iris-versicolor	0	0.974	0.026	

4. K-Means Clustering



Description

The dataset is partitioned before applying the K-Means algorithm to group similar records into clusters. The Cluster Assigner assigns each record to a cluster, while the Shape Manager and Scatter Plot nodes are used to visualize the clustered data. This workflow demonstrates unsupervised clustering in KNIME.

Outputs:

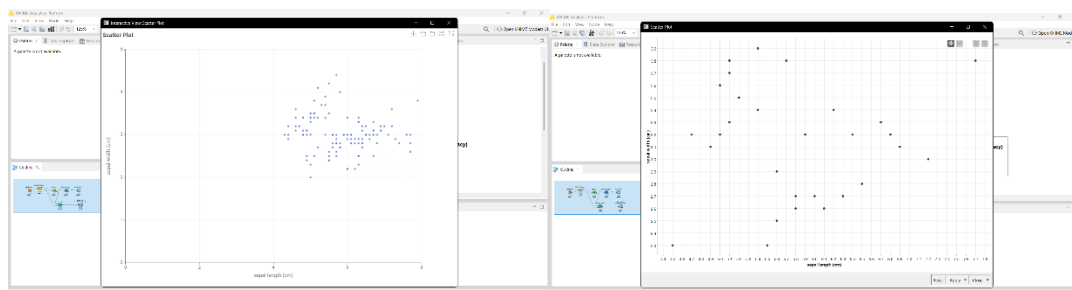


Table View - 68 - Data to Report (BIRT)							
File Edit Hilitte Navigation View							
Row ID	D sepal le...	D sepal wi...	D petal le...	D petal wi...	S target	S Cluster	
Row4	5	3.6	1.4	0.2	Iris-setosa	cluster_1	
Row9	4.9	3.1	1.5	0.1	Iris-setosa	cluster_1	
Row16	5.4	3.9	1.3	0.4	Iris-setosa	cluster_1	
Row18	5.7	3.8	1.7	0.3	Iris-setosa	cluster_1	
Row19	5.1	3.8	1.5	0.3	Iris-setosa	cluster_1	
Row21	5.1	3.7	1.5	0.4	Iris-setosa	cluster_1	
Row23	5.1	3.3	1.7	0.5	Iris-setosa	cluster_1	
Row27	5.2	3.5	1.5	0.2	Iris-setosa	cluster_1	
Row29	4.7	3.2	1.6	0.2	Iris-setosa	cluster_1	
Row31	5.4	3.4	1.5	0.4	Iris-setosa	cluster_1	
Row35	5	3.2	1.2	0.2	Iris-setosa	cluster_1	
Row41	4.5	2.3	1.3	0.3	Iris-setosa	cluster_1	
Row44	5.1	3.8	1.9	0.4	Iris-setosa	cluster_1	
Row53	5.5	2.3	4	1.3	Iris-versicolor	cluster_0	
Row54	6.5	2.8	4.6	1.5	Iris-versicolor	cluster_0	
Row64	5.6	2.9	3.6	1.3	Iris-versicolor	cluster_0	
Row67	5.8	2.7	4.1	1	Iris-versicolor	cluster_0	
Row69	5.6	2.5	3.9	1.1	Iris-versicolor	cluster_0	
Row70	5.9	3.2	4.8	1.8	Iris-versicolor	cluster_0	
Row83	6	2.7	5.1	1.6	Iris-versicolor	cluster_0	
Row92	5.8	2.6	4	1.2	Iris-versicolor	cluster_0	
Row115	6.4	3.2	5.3	2.3	Iris-virginica	cluster_2	
Row117	7.7	3.8	6.7	2.2	Iris-virginica	cluster_2	
Row123	6.3	2.7	4.9	1.8	Iris-virginica	cluster_0	
Row124	6.7	3.3	5.7	2.1	Iris-virginica	cluster_2	

Conclusion

The KNIME workflows were successfully executed and documented in a structured report. The workflows demonstrated data visualization, data manipulation, decision tree classification, and K-Means clustering. The report provides a clear overview of each workflow and its purpose.